5 Two point charges, A and B, are separated in a vacuum by a distance of 0.0100 cm, as shown in Fig. 5.1.

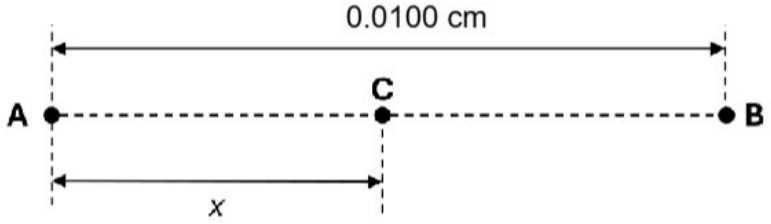
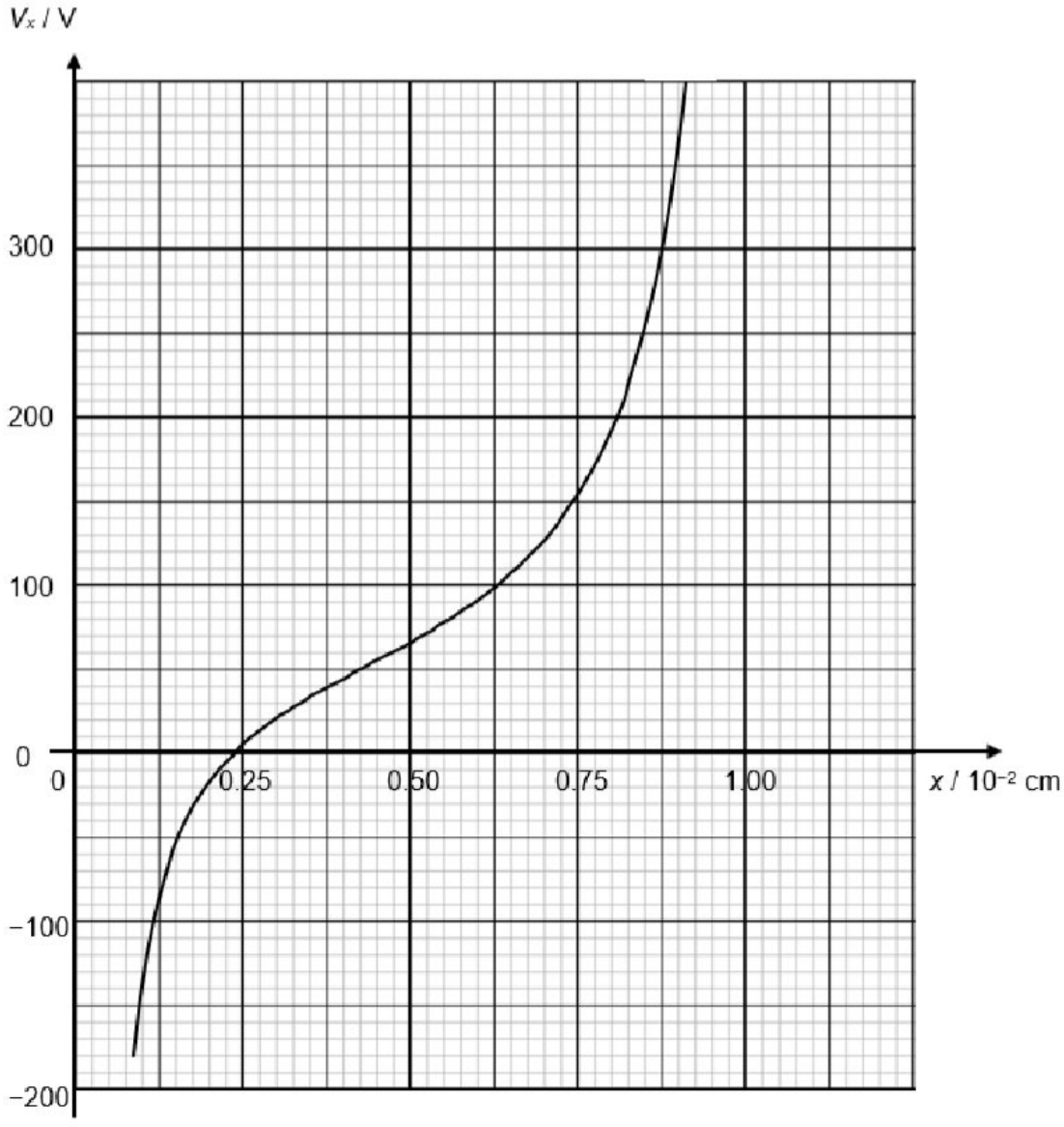


Fig. 5.1

A point C is on the line joining the two charges and is a distance x from charge A.

The variation with distance x of the electric potential V_x at point C is shown in Fig. 5.2.



(a) Define electric field strength at a point.

.....

.....

..... [1]

(b) Explain whether the charges have the same or opposite signs.

.....

.....

.....

.....

..... [2]

(c) Calculate the electric field strength where $x = 0.250 \times 10^{-2}$ cm.

electric field strength = ... V m^{-1}

direction of electric field strength: ..... [3]

(d) An electron is projected with a speed of 7.73×10^{6} m s^{-1} from the position where $x = 0.725 \times 10^{-2}$ cm towards point charge A.

Calculate the speed of the electron at the position where $x = 0.175 \times 10^{-2}$ cm.

speed = ... m s^{-1} [3]

[Total: 9]